# MSG No. 191.237 $P6'/mmm'$ [ Type III, hexagonal ]

Table 1: Wyckoff site: `1a`, site symmetry: `6'/mmm'`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24] |

Table 2: Wyckoff site: `1b`, site symmetry: `6'/mmm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24] |

Table 3: Wyckoff site: `2c`, site symmetry: `-6m2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, 0\right]$ | [1,2,3,4,5,6,7,8,9,10,11,12] |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, 0\right]$ | [13,14,15,16,17,18,19,20,21,22,23,24] |

Table 4: Wyckoff site: `2d`, site symmetry: `-6m2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$ | [1,2,3,4,5,6,7,8,9,10,11,12] |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$ | [13,14,15,16,17,18,19,20,21,22,23,24] |

Table 5: Wyckoff site: `2e`, site symmetry: `6'mm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | [1,2,3,10,11,12,13,14,15,22,23,24] |
| 2 | $[0, 0, -z]$ | [4,5,6,7,8,9,16,17,18,19,20,21] |

Table 6: Wyckoff site: `3f`, site symmetry: `mmm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | [1,5,8,10,14,16,19,23] |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | [2,6,9,11,15,17,20,24] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [3,4,7,12,13,18,21,22] |

Table 7: Wyckoff site: `3g`, site symmetry: `mmm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [1,5,8,10,14,16,19,23] |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [2,6,9,11,15,17,20,24] |

| | | |
|---|---|---|
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[3,4,7,12,13,18,21,22]` |

Table 8: Wyckoff site: `4h`, site symmetry: `3m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, z\right]$ | `[1,2,3,10,11,12]` |
| 2 | $\left[\frac{1}{3}, \frac{2}{3}, -z\right]$ | `[4,5,6,7,8,9]` |
| 3 | $\left[\frac{2}{3}, \frac{1}{3}, z\right]$ | `[13,14,15,22,23,24]` |
| 4 | $\left[\frac{2}{3}, \frac{1}{3}, -z\right]$ | `[16,17,18,19,20,21]` |

Table 9: Wyckoff site: `6i`, site symmetry: `2'mm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, z\right]$ | `[1,10,14,23]` |
| 2 | $\left[0, \frac{1}{2}, z\right]$ | `[2,11,15,24]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[3,12,13,22]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, -z\right]$ | `[4,7,18,21]` |
| 5 | $\left[\frac{1}{2}, 0, -z\right]$ | `[5,8,16,19]` |
| 6 | $\left[0, \frac{1}{2}, -z\right]$ | `[6,9,17,20]` |

Table 10: Wyckoff site: `6j`, site symmetry: `m2'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | `[1,8,16,23]` |
| 2 | $[0, x, 0]$ | `[2,9,17,24]` |
| 3 | $[-x, -x, 0]$ | `[3,7,18,22]` |
| 4 | $[x, x, 0]$ | `[4,12,13,21]` |
| 5 | $[-x, 0, 0]$ | `[5,10,14,19]` |
| 6 | $[0, -x, 0]$ | `[6,11,15,20]` |

Table 11: Wyckoff site: `6k`, site symmetry: `m2'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, 0, \frac{1}{2}\right]$ | `[1,8,16,23]` |
| 2 | $\left[0, x, \frac{1}{2}\right]$ | `[2,9,17,24]` |
| 3 | $\left[-x, -x, \frac{1}{2}\right]$ | `[3,7,18,22]` |
| 4 | $\left[x, x, \frac{1}{2}\right]$ | `[4,12,13,21]` |
| 5 | $\left[-x, 0, \frac{1}{2}\right]$ | `[5,10,14,19]` |
| 6 | $\left[0, -x, \frac{1}{2}\right]$ | `[6,11,15,20]` |

Table 12: Wyckoff site: 6l, site symmetry: mm2

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 2x,\ 0]$ | [1,5,8,10] |
| 2 | $[-2x,\ -x,\ 0]$ | [2,6,9,11] |
| 3 | $[x,\ -x,\ 0]$ | [3,4,7,12] |
| 4 | $[-x,\ x,\ 0]$ | [13,18,21,22] |
| 5 | $[-x,\ -2x,\ 0]$ | [14,16,19,23] |
| 6 | $[2x,\ x,\ 0]$ | [15,17,20,24] |

Table 13: Wyckoff site: 6m, site symmetry: mm2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 2x,\ \frac{1}{2}\right]$ | [1,5,8,10] |
| 2 | $\left[-2x,\ -x,\ \frac{1}{2}\right]$ | [2,6,9,11] |
| 3 | $\left[x,\ -x,\ \frac{1}{2}\right]$ | [3,4,7,12] |
| 4 | $\left[-x,\ x,\ \frac{1}{2}\right]$ | [13,18,21,22] |
| 5 | $\left[-x,\ -2x,\ \frac{1}{2}\right]$ | [14,16,19,23] |
| 6 | $\left[2x,\ x,\ \frac{1}{2}\right]$ | [15,17,20,24] |

Table 14: Wyckoff site: 12n, site symmetry: ..m'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ z]$ | [1,23] |
| 2 | $[0,\ x,\ z]$ | [2,24] |
| 3 | $[-x,\ -x,\ z]$ | [3,22] |
| 4 | $[x,\ x,\ -z]$ | [4,21] |
| 5 | $[-x,\ 0,\ -z]$ | [5,19] |
| 6 | $[0,\ -x,\ -z]$ | [6,20] |
| 7 | $[-x,\ -x,\ -z]$ | [7,18] |
| 8 | $[x,\ 0,\ -z]$ | [8,16] |
| 9 | $[0,\ x,\ -z]$ | [9,17] |
| 10 | $[-x,\ 0,\ z]$ | [10,14] |
| 11 | $[0,\ -x,\ z]$ | [11,15] |
| 12 | $[x,\ x,\ z]$ | [12,13] |

Table 15: Wyckoff site: 12o, site symmetry: .m.

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 2x,\ z]$ | [1,10] |
| 2 | $[-2x,\ -x,\ z]$ | [2,11] |
| 3 | $[x,\ -x,\ z]$ | [3,12] |
| 4 | $[x,\ -x,\ -z]$ | [4,7] |
| 5 | $[x,\ 2x,\ -z]$ | [5,8] |
| 6 | $[-2x,\ -x,\ -z]$ | [6,9] |
| 7 | $[-x,\ x,\ z]$ | [13,22] |

*continued ...*

Table 15

| No. | position | mapping |
|---|---|---|
| 8 | $[-x,\ -2x,\ z]$ | [14,23] |
| 9 | $[2x,\ x,\ z]$ | [15,24] |
| 10 | $[-x,\ -2x,\ -z]$ | [16,19] |
| 11 | $[2x,\ x,\ -z]$ | [17,20] |
| 12 | $[-x,\ x,\ -z]$ | [18,21] |

Table 16: Wyckoff site: 12p, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | [1,8] |
| 2 | $[-y,\ x-y,\ 0]$ | [2,9] |
| 3 | $[-x+y,\ -x,\ 0]$ | [3,7] |
| 4 | $[x,\ x-y,\ 0]$ | [4,12] |
| 5 | $[-x+y,\ y,\ 0]$ | [5,10] |
| 6 | $[-y,\ -x,\ 0]$ | [6,11] |
| 7 | $[x-y,\ x,\ 0]$ | [13,21] |
| 8 | $[-x,\ -y,\ 0]$ | [14,19] |
| 9 | $[y,\ -x+y,\ 0]$ | [15,20] |
| 10 | $[x-y,\ -y,\ 0]$ | [16,23] |
| 11 | $[y,\ x,\ 0]$ | [17,24] |
| 12 | $[-x,\ -x+y,\ 0]$ | [18,22] |

Table 17: Wyckoff site: 12q, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ y,\ \frac{1}{2}\right]$ | [1,8] |
| 2 | $\left[-y,\ x-y,\ \frac{1}{2}\right]$ | [2,9] |
| 3 | $\left[-x+y,\ -x,\ \frac{1}{2}\right]$ | [3,7] |
| 4 | $\left[x,\ x-y,\ \frac{1}{2}\right]$ | [4,12] |
| 5 | $\left[-x+y,\ y,\ \frac{1}{2}\right]$ | [5,10] |
| 6 | $\left[-y,\ -x,\ \frac{1}{2}\right]$ | [6,11] |
| 7 | $\left[x-y,\ x,\ \frac{1}{2}\right]$ | [13,21] |
| 8 | $\left[-x,\ -y,\ \frac{1}{2}\right]$ | [14,19] |
| 9 | $\left[y,\ -x+y,\ \frac{1}{2}\right]$ | [15,20] |
| 10 | $\left[x-y,\ -y,\ \frac{1}{2}\right]$ | [16,23] |
| 11 | $\left[y,\ x,\ \frac{1}{2}\right]$ | [17,24] |
| 12 | $\left[-x,\ -x+y,\ \frac{1}{2}\right]$ | [18,22] |

Table 18: Wyckoff site: 24r, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $[-y,\ x-y,\ z]$ | [2] |

*continued ...*

Table 18

| No. | position | mapping |
|---|---|---|
| 3 | $[-x+y,\ -x,\ z]$ | [3] |
| 4 | $[x,\ x-y,\ -z]$ | [4] |
| 5 | $[-x+y,\ y,\ -z]$ | [5] |
| 6 | $[-y,\ -x,\ -z]$ | [6] |
| 7 | $[-x+y,\ -x,\ -z]$ | [7] |
| 8 | $[x,\ y,\ -z]$ | [8] |
| 9 | $[-y,\ x-y,\ -z]$ | [9] |
| 10 | $[-x+y,\ y,\ z]$ | [10] |
| 11 | $[-y,\ -x,\ z]$ | [11] |
| 12 | $[x,\ x-y,\ z]$ | [12] |
| 13 | $[x-y,\ x,\ z]$ | [13] |
| 14 | $[-x,\ -y,\ z]$ | [14] |
| 15 | $[y,\ -x+y,\ z]$ | [15] |
| 16 | $[x-y,\ -y,\ -z]$ | [16] |
| 17 | $[y,\ x,\ -z]$ | [17] |
| 18 | $[-x,\ -x+y,\ -z]$ | [18] |
| 19 | $[-x,\ -y,\ -z]$ | [19] |
| 20 | $[y,\ -x+y,\ -z]$ | [20] |
| 21 | $[x-y,\ x,\ -z]$ | [21] |
| 22 | $[-x,\ -x+y,\ z]$ | [22] |
| 23 | $[x-y,\ -y,\ z]$ | [23] |
| 24 | $[y,\ x,\ z]$ | [24] |